AI 683 – Statistical Learning

Kaggle Competition Project: Playground Series – Season 5, Episode 11 Predicting Loan Payback

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Problem and Data

The task is a **binary classification problem**: predict whether a loan will be paid back.

- **Dataset**: Tabular data with applicant income, co-applicant income, loan amount, credit score, and categorical attributes.
- test.csv, train.csv, submission.csv dataset provided.
- **Target variable**: Loan status (paid back vs. default).
- **Evaluation metric**: ROC–AUC, as specified by the competition.

Methods

- **Preprocessing**:
  - Numeric features → median imputation + scaling.
  - Categorical features → most-frequent imputation + one-hot encoding.
- **Feature engineering**:
  - total_income (applicant + co-applicant).
  - loan_income_ratio (loan amount / total income).
  - Log transforms for skewed variables.
  - Credit score buckets.
- **Models**:
  - Baseline: Logistic Regression.
  - Tree-based: Random Forest, LightGBM, XGBoost, CatBoost.
  - Ensemble: Logistic Regression meta-model stacking LGBM, XGB, and CatBoost.
- **Validation**: Stratified 5-fold cross-validation for robust OOF estimates.

- **Tuning**: Grid search for Logistic Regression; tuned boosting parameters for LGBM/XGB/CatBoost.

Results

- **Base model OOF AUCs**:
    - LightGBM: 0.92197
    - XGBoost: 0.92145
    - CatBoost: 0.92147

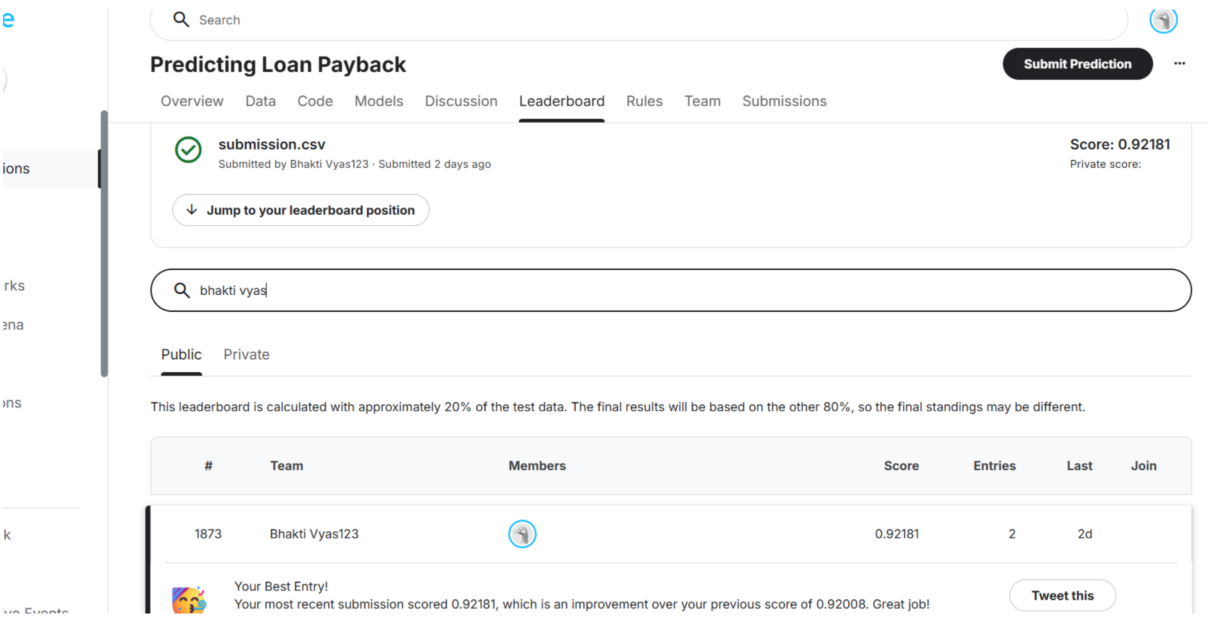
OOF AUC: **0.9223**

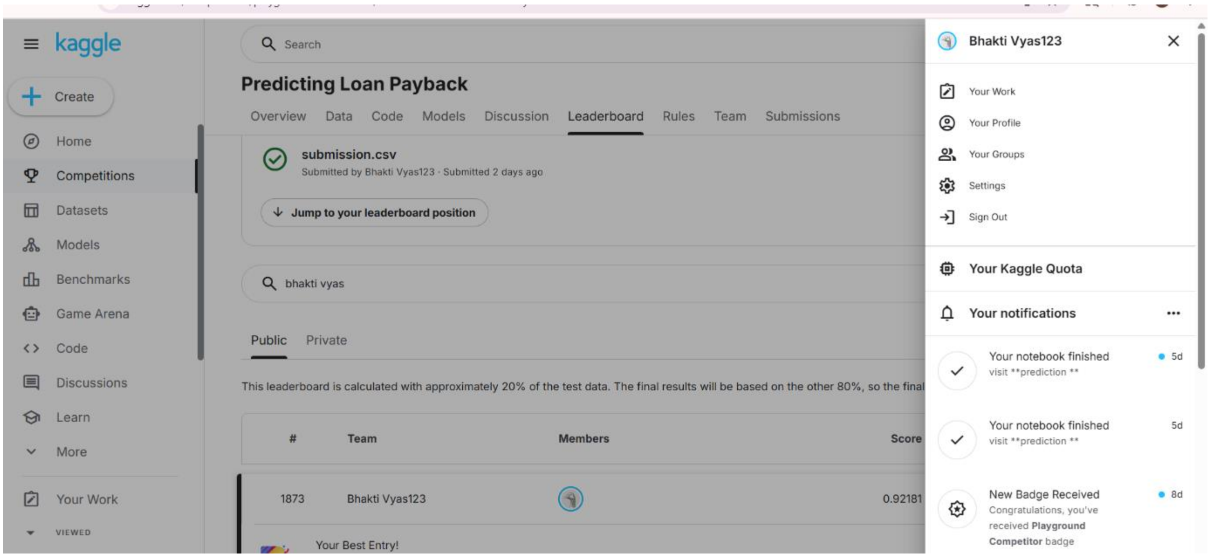
Leaderboard

- Previous best: 0.92008
- Current best: 0.92181

Conclusion

Using systematic preprocessing, feature engineering, and model comparison, we obtained high and stable scores on the loan payback prediction exercise. As the individual gradient boosting models had reached an OOF ROC -AUC of approximately 0.9215 a stack of CatBoost, LightGBM and XGBoost with a Logistic Regression meta-model boosted the ensemble to 0.9223 OOF AUC.

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